

Gradient Boost Tree Network based on Extensive Feature Analysis for Popularity Prediction of Social Posts

Chih-Chung Hsu

Institute of Data Science, National Cheng Kung University
Tainan, Taiwan
cchsu@gs.ncku.edu.tw

Xiu-Yu Hou

Institute of Data Science, National Cheng Kung University
Tainan, Taiwan
xiuyu.hou.tw@gmail.com

Chia-Ming Lee

Institute of Data Science, National Cheng Kung University
Tainan, Taiwan
zuw408421476@gmail.com

Chi-Han Tsai

Institute of Data Science, National Cheng Kung University
Tainan, Taiwan
fateplsf567@gmail.com

ABSTRACT

Social media popularity (SMP) prediction is a complex task, affected by various features such as text, images, and spatial-temporal information. One major challenge in SMP is integrating features from multiple modalities without overemphasizing user-specific details while efficiently capturing relevant user information. This study introduces a robust multi-modality feature mining framework for predicting SMP scores by incorporating additional identity-related features sourced from the official SMP dataset when a user's path alias is accessible. Our preliminary analyses suggest these supplemental features significantly enrich the user-related context, contributing to a substantial improvement in performance and proving that non-identity features are relatively unimportant. This implies that we should focus more on discovering the identity-related features than other meta-data. To further validate our findings, we perform comprehensive experiments investigating the relationship between those identity-related features and scores. Finally, the LightGBM and TabNet are employed within our framework to effectively capture intricate semantic relationships among different modality features and user-specific data. Our experimental results confirm that these identity-related features, especially external ones, significantly improve the prediction performance of SMP tasks.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks; Classification and regression trees.**

This research was supported in part by Higher Education Sprout Project and High Education Humanities and Social Science Subjects Benchmark Project, Ministry of Education to the Headquarters of University Advancement at National Cheng Kung University(NCKU).

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MM '23, October 29-November 3, 2023, Ottawa, ON, Canada

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0108-5/23/10...\$15.00
<https://doi.org/10.1145/3581783.3612843>

KEYWORDS

Time-series, Multi-modal, User-profile feature, LightGBM, TabNet

ACM Reference Format:

Chih-Chung Hsu, Chia-Ming Lee, Xiu-Yu Hou, and Chi-Han Tsai. 2023. Gradient Boost Tree Network based on Extensive Feature Analysis for Popularity Prediction of Social Posts. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29-November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3581783.3612843>

1 INTRODUCTION

In recent years, people have preferred using social media to share their daily lives and engage in marketing activities, thereby leveraging their personal or brand influence. In other words, social media platforms hold immense value. Consequently, the accurate analysis of social media and user behavior has emerged as a burgeoning research topic.

Social media platforms encompass various types of data, posing challenges in accurately predicting SMP tasks due to the complexity of feature combinations. Appropriate feature transformations are critical for handling different feature types. Previous studies have been limited by weak feature extractors, hindering the improvement of task baselines.

In methodologies development, several prior arts have been proposed in recent years for SMP tasks. The peer methods in [6, 7, 9] models treat the prediction task in SMP as a multi-modal regression task. For feature extraction, conventional natural language processing (NLP) models are typically employed to obtain semantic features from text information, such as RoBERTa [13] and Sentence-Transformer [16]. In [6], it is claimed that incorporating image features and generative captions can further improve the performance of the SMP task. Some recent state-of-the-art, such as [24][2][19], have integrated multi-modal features and external user information, reaching outstanding performance in SMP tasks. However, lacking comprehensive study on the impact of external identity-related features results in difficulty in further improving performance. We, therefore, provide a complete discussion and experiments on discovering how the identity-related features important is.

Specifically, HyFea[11], proposed by Lai et al., highlights that we can obtain additional user profile data, such as totalViews (#counts of viewers) and totalFollowing (How many users follow the targeted user). Mining additional features correlated with the user (identity)

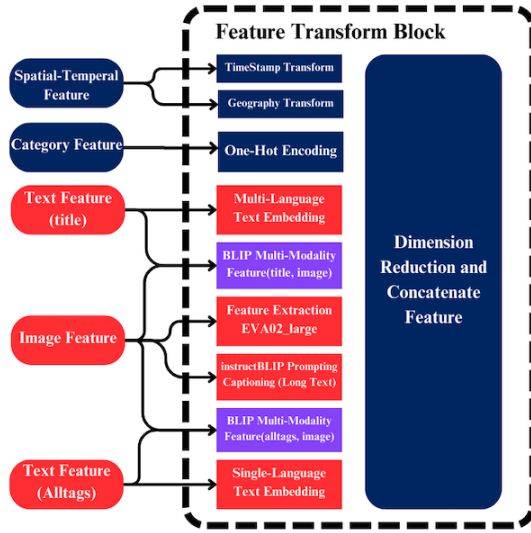


Figure 1: The illustrations for the multi-modal feature transformation module(MFTM). Features are processed by specific transformation, followed by dimension reduction(PCA), and are then concatenated together.

can enrich user information, thereby improving the performance of SMP tasks. We conduct a series of experiments to investigate its correlation. Specifically, the external identity-related feature reflects the characteristics of an "internet celebrity." Posts from internet celebrities are often more likely to attract attention, thereby affecting the popularity of their posts.

However, in traditional SMP tasks [21–23], data was typically split into training and testing sets based solely on the timestamp. This approach needed to account for potential user differences between the sets. As user identity plays a crucial role in SMP tasks [6–9], improper data splitting methods could result in missing UIDs and consequently limit the performance of the SMP task. On the one hand, the profile-related features significantly contribute to the performance of SMP tasks. On the other hand, overly relying on such identity-related features could result in bad prediction results when its identity is new. Therefore, to avoid overly focusing on user-specific details and promote a more generalized approach in SMP tasks, we utilize group k-fold validation based on the number of unique UIDs, ensuring non-overlapping users in each fold. This allows us to retain crucial user information while preventing the model from over-relying on user-specific details, striking a balance between capturing semantic information and maintaining generalizability.

To handle popularity prediction using features extracted from social media, traditional discrete regression schemes have been employed, including XGBoost (XGB) [1], Random Forest (RF) [12], LightGBM (LGBM) [10], CatBoost (CATB) [15], and even model stacking [14]. These supervised learning and ensemble-based strategies have demonstrated their effectiveness across various tasks, particularly in tabular data.

To earn better performance of SMP tasks, as well as fully discover the identity-related, this paper proposes a robust and reliable

time-series regression framework based on multi-modal feature aggregation and external profile features mining based on extensive feature analysis.

Firstly, we improve the previously proposed multi-modal feature aggregation module proposed in [8] last year by replacing it with more advantageous feature extractors. Additionally, we apply a group k-fold validation that preserves user information for prediction tasks. This setting is essential to truly and fairly evaluate the performance during the training phase. Secondly, we compare the impact of different identity-related features and other features from meta-data on popularity prediction. We analyze whether the semantic meaning of external features aligns with the semantic meaning of popularity. The contribution of this paper is threefold:

- We adjust the feature transformation process and compare it with the existing module to investigate the impact of different feature combinations on the model's performance.
- To better observe the effect of stacking multi-modality feature, We conduct some experiments to make the ablation study to examine the influence of the different semantic representations.
- We discover that external identity-related features have a significant impact on the SMP task. Therefore, we conduct comprehensive experiments to observe the relationship between different identity-related features and the popularity score.

The rest of this paper is organized as follows. Section 2 presents the proposed multi-modal feature transformation and the robust prediction framework for the SMP task. Section 3 shows our experimental results. Finally, conclusions are drawn in Section 4.

2 ADVANCED ANALYSIS OF MULTI-MODAL FEATURE TRANSFORMATION

2.1 Overview

Building upon prior research, the SMP dataset houses numerous distinctive features categorized into six domains: numerical, categorical, temporal, spatial, text, and image. A key step after data extraction involves an elaborate feature transformation module to convert each feature into different-dimension embedding vectors or variable forms for a comprehensive analysis. This is visually represented in Fig. 1. Our comprehensive analysis demonstrates that, contrary to prior assumptions, the most critical feature is profile-based, thus warranting external feature crawling.

2.2 Detailed Feature Transform Block

The SMP dataset exhibits considerable complexity and variability across different feature types. To navigate this complexity, we carry out targeted feature transformations, deriving diverse embedding vectors or alternative features. These transformations culminate in a comprehensive representation of a social media post.

Image Data Analysis. Previous work [6] has shown that image captioning offers significant benefits for SMP tasks. Using this insight, our study leverages the instructBLIP [3] model in tandem with prompt engineering ("Write a detailed description") to generate descriptive image captions averaging 400 words. A pre-trained Sentence Transformer [16] extracts semantic features from these

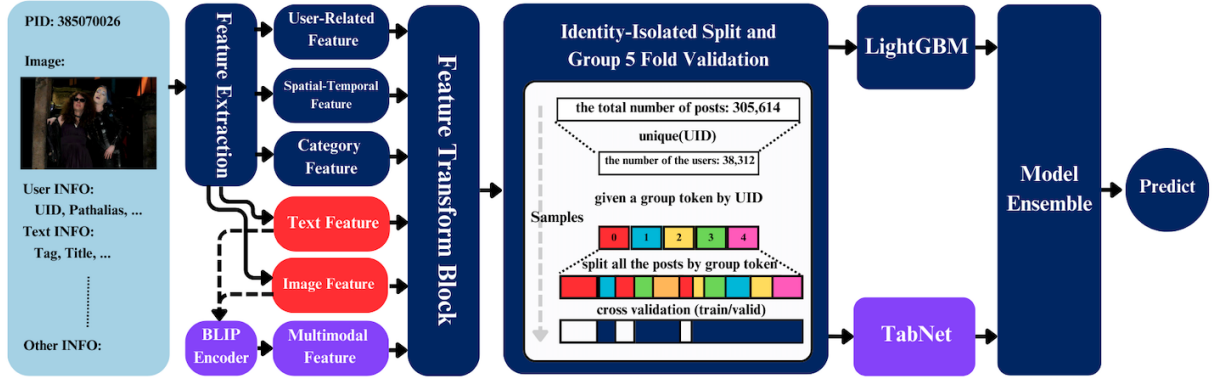


Figure 2: Flowchart of the proposed regression model with multi-modal feature mining.

captions (f_{cap}). Simultaneously, a pre-trained EVA02-large [4, 20] model coupled with PCA [5] is used for image feature extraction and dimensionality reduction (f_{image}).

Numerical Data Analysis. The UID and post-ID (PID) are converted into unique integers, while the post publish timestamp is transformed into a DateTime variable (f_{time}). Spatial data such as latitude and longitude are encoded via Geocoder, transforming them into categorical variables (city, state, country) represented as f_{geo} . We utilize HyFea[11] to get external user information data, denoted as f_{eu} . All external user information data is connected to UID. In other words, these features are identity-related. The remainder of the numerical data is denoted as f_n .

Text-based Data Analysis. Our analysis uncovers that textual features possess multilingual attributes. To prevent performance degradation caused by single language-trained feature extractors, we use Sentence Transformers [16, 17] and PCA to derive $f_{multi-lang}$ and $f_{single-lang}$ representations.

Categorical Data Analysis. Categorical data are represented as one-hot vector representations, denoted as f_{cat} .

Multi-Modality Data Analysis. Since SMP dataset does not allow direct access to multi-modal features, we combine features across different modalities for extraction. Observations reveal non-overlapping semantic information between various text features. This realization leads us to pair different text types with images, generating distinct output vectors that are subsequently dimensionally reduced using PCA, resulting in the final feature representation (f_m).

Feature Aggregation. Once all pertinent information is processed, we aggregate all features as $f_{all} = [f_{cap}, f_{image}, f_{time}, f_{geo}, f_n, f_{eu}, f_{single-lang}, f_{multi-lang}, f_{cat}, f_m]$ serving as the basis for social media popularity prediction.

2.3 Group k-fold Validation and Prediction for Greater Robustness

In the context of social media prediction, user-specific information significantly impacts the results. However, considering the possible lack of continuity in user presence across training and testing sets, our objective is to develop a model that can still capture the dataset's

essence. Our chosen approach involves a unique group k-fold cross-validation where UIDs are divided into k groups. This strategy increases model stability and curtails overfitting. In each fold, the model predicts the testing set's popularity. The median value of the predicted popularity scores from all folds is used for the final prediction, enhancing prediction reliability and model robustness.

2.4 Model Ensemble Strategy for Enhanced Predictive Power

A model ensemble leverages the predictions of multiple independent models to augment overall predictive performance and stability. This study benefits from the synergy of LightGBM [10] - known for its proficiency in managing tabular data, and TabNet [18] - known for handling high-dimensional complex tabular data. The final prediction of social media popularity is obtained by computing a weighted average of the predictions from these two models. This ensemble strategy enhances the robustness of our model, reinforcing the importance of our findings on the critical role of profile-based features in social media popularity prediction.

3 EXPERIMENTAL RESULTS

The SMP Dataset (SMPD), derived from [21–23], contains 305,614 posts with an array of diverse data types. As per [9], the treatment of missing data is specific. For performance evaluation, the study utilizes Spearman's Rho rank correlation (SRC)[25] and Mean Absolute Error (MAE) as evaluation metrics.

3.1 Performance Evaluation of the Proposed Method

This section details the performance assessment of the traditional SMP task executed via the proposed multi-modal feature mining framework.

As delineated in Table 1, our strategy optimizes the prior model using findings from the 2022 SMP competition. We enrich the model by stacking external identity-related features and employing advanced feature extractors, thereby significantly outperforming the preceding model[8].

Our experimental findings accentuate that supplementary identity-related features such as totalFaves, followers, and totalInGroup, exert the most substantial impact on model performance. Contrarily, other features present a comparatively restrained effect on performance, although contributing incrementally to the final result. Besides LightGBM, suitable for tabular data analysis, we integrate TabNet due to its prowess in analyzing complex semantic representations within embedding vectors. The culmination of these strategies in our model ensemble yields an SRC of **0.718** and an MAE of **1.241** for our final submission in the 2023 ACM Multimedia Social Media Prediction Challenge.

Table 1: Performance comparison of proposed method based on stacking different features.

	LGBM		TabNet	
	SRC ↑	MAE ↓	SRC ↑	MAE ↓
The result of the SMP in 2022	0.636	1.436	0.625	1.502
+ External Identity-related Feature	0.707	1.261	0.66	1.362
+ Image and Text Feature	0.708	1.26	0.668	1.366
+ Multi-modality Feature	0.711	1.252	0.671	1.354
+ Image Captioning Feature	0.713	1.249	0.682	1.345

3.2 Ablation Study of MFTM for SMP Task

We acknowledge the potential for different features to variably influence model performance. In an effort to comprehend this, we orchestrate ablation experiments to explore the effects of distinct features, as tabulated in Table 2. When utilizing image, text, and multi-modality features separately, performance is roughly equivalent. If a singular feature extractor is sufficiently potent and not fine-tuned, it is feasible to train the model solely on a single modality feature. This approach mitigates computational complexity and memory usage. However, when computational resources are copious, stacking all features potentially augments performance.

3.3 Comparison of the Influence of External Features

Upon contrasting the correlation coefficients of these external features with the popularity score against those of the original features, we discern a more robust correlation in the former, as shown in Table 3. This evidence suggests the identity-related features' pivotal role in predicting a post's popularity score. Further, we generate visualizations in Figure 3, projecting user profile features strongly

Table 2: Ablation study about multi-modality feature transformation module for the social media prediction task.

Userprofile	Image	Text	Multimodal	Captioning	SRC ↑	MAE ↓
✓	✓	✓	✓	✓	0.65	1.41
✓	✓				0.708	1.258
✓		✓			0.708	1.26
✓	✓	✓			0.712	1.251
✓			✓	✓	0.709	1.257
✓	✓	✓	✓		0.711	1.252
✓	✓	✓		✓	0.711	1.25
✓	✓	✓	✓	✓	0.713	1.249

Ranking	Variable	SRC(abs) ↑
top1	Popularity score	1
top2	follower	0.376
top3	totalInGroup	0.31
top4	totalViews	0.30
top5	following	0.29
top6	totalFaves	0.28
top7	totalTags	0.21
top8	totalGeotagged	0.16
top9	totalImages	0.03
	Average of external features	0.257
	Average of original features	0.012

Table 3: Spearman's Rho rank Correlation Coefficient with Popularity score between external identity-related features and original numeric features.

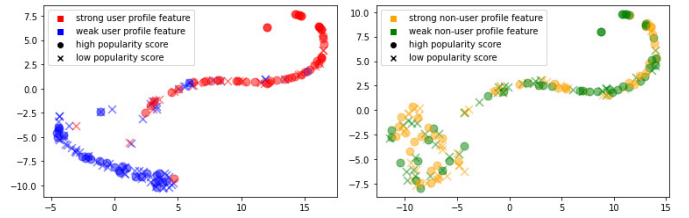


Figure 3: The t-SNE visualization. The left figure shows a link between strong user profile features and popularity. The right figure shows that non-user profile features don't noticeably affect popularity

correlating with popularity score alongside arbitrarily selected non-user profile features onto a two-dimensional plane. These visualizations depict a semantic association between user-profile features and the popularity score. In our experimental parameters, a "strong" user profile denotes a more extensive and detailed profile, while a "weak" profile suggests a less comprehensive user profile.

We observe that specific external identity-related features, indicative of user popularity and social connections within the community, wield a particularly high influence. These findings underscore the paramount importance of these external features in the determination of a post's popularity score.

4 CONCLUSIONS

This paper introduces a robust multi-modality feature mining framework that amalgamates different features to capture a comprehensive range of semantic information. This comprehensive approach yields more stable predictions for social media popularity scores. The experiment substantiates the premise that **"the stronger the user profile, the more popular the posts tend to be."** We postulate that the degree of user interconnection, indicative of "Internet celebrity" behaviors, bears more influence than other features. Moreover, when contrasting different features, we ascertain that integrating more multi-modal features also provides a tangible benefit for the SMP task.

REFERENCES

- [1] Tianqi Chen and Carlos Guestrin. 2016. Xgboost: A scalable tree boosting system. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 785–794.
- [2] Weilong Chen, Chenghao Huang, Weimin Yuan, Xiaolu Chen, Wenhao Hu, Xinran Zhang, and Yanru Zhang. 2022. Title-and-Tag Contrastive Vision-and-Language Transformer for Social Media Popularity Prediction. In *Proceedings of the 30th ACM International Conference on Multimedia* (Lisboa, Portugal) (MM '22). Association for Computing Machinery, New York, NY, USA, 7008–7012. <https://doi.org/10.1145/3503161.3551568>
- [3] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. 2023. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. *arXiv:2305.06500* [cs.CV]
- [4] Yuxin Fang, Quan Sun, Xinggong Wang, Tiejun Huang, Xinlong Wang, and Yue Cao. 2023. EVA-02: A Visual Representation for Neon Genesis. *arXiv preprint arXiv:2303.11331* (2023).
- [5] Karl Pearson F.R.S. 1901. LIII. On lines and planes of closest fit to systems of points in space. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 2, 11 (1901), 559–572. <https://doi.org/10.1080/14786440109462720>
- [6] Chih-Chung Hsu, Li-Wei Kang, Chia-Yen Lee, Jun-Yi Lee, Zhong-Xuan Zhang, and Shao-Min Wu. 2019. Popularity Prediction of Social Media Based on Multi-Modal Feature Mining. In *Proceedings of the 27th ACM International Conference on Multimedia* (Nice, France) (MM '19). Association for Computing Machinery, New York, NY, USA, 2687–2691. <https://doi.org/10.1145/3343031.3356064>
- [7] Chih-Chung Hsu, Chia-Yen Lee, Ting-Xuan Liao, Jun-Yi Lee, Tsai-Yne Hou, Ying-Chu Kuo, Jing-Wen Lin, Ching-Yi Hsueh, Zhong-Xuan Zhang, and Hsiang-Chin Chien. 2018. An iterative refinement approach for social media headline prediction. In *Proceedings of the ACM International Conference on Multimedia*. ACM, 2008–2012.
- [8] Chih-Chung Hsu, Pi-Ju Tsai, Ting-Chun Yeh, and Xiu-Yu Hou. 2022. A Comprehensive Study of Spatiotemporal Feature Learning for Social Media Popularity Prediction. In *Proceedings of the 30th ACM International Conference on Multimedia* (Lisboa, Portugal) (MM '22). Association for Computing Machinery, New York, NY, USA, 7130–7134. <https://doi.org/10.1145/3503161.3551593>
- [9] Chih-Chung Hsu, Wen-Hai Tseng, Hao-Ting Yang, Chia-Hsiang Lin, and Chih-Hung Kao. 2020. Rethinking Relation between Model Stacking and Recurrent Neural Networks for Social Media Prediction. In *Proceedings of the 28th ACM International Conference on Multimedia*. Association for Computing Machinery, New York, NY, USA, 4585–4589. <https://doi.org/10.1145/3394171.3417332>
- [10] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*. 3146–3154.
- [11] Xin Lai, Yihong Zhang, and Wei Zhang. 2020. HyFea: Winning Solution to Social Media Popularity Prediction for Multimedia Grand Challenge 2020. In *Proceedings of the 28th ACM International Conference on Multimedia* (Seattle, WA, USA) (MM '20). Association for Computing Machinery, New York, NY, USA, 4565–4569. <https://doi.org/10.1145/3394171.3416273>
- [12] Andy Liaw, Matthew Wiener, et al. 2002. Classification and regression by randomForest. *R News* 2, 3 (2002), 18–22.
- [13] Yinhan Liu, Mye Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692* (2019).
- [14] Kevin Matlock, Carlos De Niz, Raziur Rahman, Souparno Ghosh, and Ranadip Pal. 2018. Investigation of model stacking for drug sensitivity prediction. *BMC bioinformatics* 19, 3 (2018), 21–33.
- [15] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. 2018. CatBoost: unbiased boosting with categorical features. *Advances in neural information processing systems* 31 (2018).
- [16] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. <https://arxiv.org/abs/1908.10084>
- [17] Nils Reimers and Iryna Gurevych. 2020. Making Monolingual Sentence Embeddings Multilingual using Knowledge Distillation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. <https://arxiv.org/abs/2004.09813>
- [18] Tomas Pfister, Serkan O. Arik. 2019. TabNet: Attentive Interpretable Tabular Learning. *arXiv:1908.07442* (2019).
- [19] YunPeng Tan, Fangyu Liu, BoWei Li, Zheng Zhang, and Bo Zhang. 2022. An Efficient Multi-View Multimodal Data Processing Framework for Social Media Popularity Prediction. In *Proceedings of the 30th ACM International Conference on Multimedia* (Lisboa, Portugal) (MM '22). Association for Computing Machinery, New York, NY, USA, 7200–7204. <https://doi.org/10.1145/3503161.3551607>
- [20] Ross Wightman. 2019. PyTorch Image Models. <https://github.com/huggingface/pytorch-image-models>. <https://doi.org/10.5281/zenodo.4414861>
- [21] Bo Wu, Wen-Huang Cheng, Peiye Liu, Bei Liu, Zhaoyang Zeng, and Jiebo Luo. 2019. SMP Challenge: An Overview of Social Media Prediction Challenge 2019. In *Proceedings of the 27th ACM International Conference on Multimedia*.
- [22] Bo Wu, Wen-Huang Cheng, Yongdong Zhang, Huang Qiushi, Li Jintao, and Tao Mei. 2017. Sequential Prediction of Social Media Popularity with Deep Temporal Context Networks. In *International Joint Conference on Artificial Intelligence (IJCAI)* (Melbourne, Australia).
- [23] Bo Wu, Tao Mei, Wen-Huang Cheng, and Yongdong Zhang. 2016. Unfolding Temporal Dynamics: Predicting Social Media Popularity Using Multi-scale Temporal Decomposition. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence (AAAI)* (Phoenix, Arizona).
- [24] Jianmin Wu, Liming Zhao, Dangwei Li, Chen-Wei Xie, Siyang Sun, and Yun Zheng. 2022. Deeply Exploit Visual and Language Information for Social Media Popularity Prediction. In *Proceedings of the 30th ACM International Conference on Multimedia* (Lisboa, Portugal) (MM '22). Association for Computing Machinery, New York, NY, USA, 7045–7049. <https://doi.org/10.1145/3503161.3551576>
- [25] Sheng Yue, Paul Pilon, and George Cavadias. 2002. Power of the Mann–Kendall and Spearman's rho tests for detecting monotonic trends in hydrological series. *Journal of hydrology* 259, 1–4 (2002), 254–271.